Applied Soft Computing 40 (2016) 221–235

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

An interactive fuzzy programming approach for bi-objective straight and U-shaped assembly line balancing problem

M.H. Alavidoost [a,*], Hossein Babazadeh [b], S.T. Sayyari [c]

[a] Department of Industrial Engineering, AmirKabir University of Technology, Tehran, Iran
[b] Department of Industrial Engineering, University of Science and Culture, Tehran, Iran
[c] Department of Industrial Engineering, University of Science and Technology, Tehran, Iran

ARTICLE INFO

Article history:
Received 10 March 2015
Received in revised form 18 October 2015
Accepted 10 November 2015
Available online 2 December 2015

Keywords:
Interactive fuzzy programming
Multi-objective optimization
Assembly line balancing
Fuzzy numbers

ABSTRACT

The consideration of this paper is given to address the straight and U-shaped assembly line balancing problem. Although many attempts in the literature have been made to develop deterministic version of the assembly line model, the attention is not considerably given to those in uncertain environment. In this paper, a novel bi-objective fuzzy mixed-integer linear programming model (BOFMILP) is developed so that triangular fuzzy numbers (TFNs) are employed in order to represent uncertainty and vagueness associated with the task processing times in the real production systems. In this proposed model, two conflicting objectives (minimizing the number of stations as well as cycle time) are considered simultaneously with respect to set of constraints. For this purpose, an appropriate strategy in which new two-phase interactive fuzzy programming approach is proposed as a solution method to find an efficient compromise solution. Finally, validity of the proposed model as well as its solution approach are evaluated though numerical examples. In addition, a comparison study is conducted over some test problems in order to assess the performance of the proposed solution approach. The results demonstrate that our proposed interactive fuzzy approach not only can be applied in ALBPs but also is capable to handle any practical MOLP models. Moreover, in light of these results, the proposed model may constitute a framework aiming to assist the decision maker (DM) to deal with uncertainty in assembly line problem.

1. Introduction

The growing global competitive market compels manufacturing organizations to engage themselves in all productivity improvement plans aiming to promote efficiency and effectiveness. In this direction, constructing an efficient assembly line is being considered as one of the most important issues in developing an assembly line [1]. Generally, assembly line balancing problem (ALBP) is considered as an assignment of the required tasks to the set of workstations to produce a product using either batch or mass production method with respect to the precedence relations among tasks and some other constraints. Thus, the main issue in ALBPs lies on design of appropriate order and sequence of the tasks [2]. ALBPs are generally classified into two main categories including simple ALBP (SALBP) and generalized ALBP (GALBP). GALBPs have some extra features such as cost goals, station parallelization, mixed-model production, etc. in comparison with SALBPs [3].

All versions of SALBPs with respect to their objectives are presented in Table 1 [4].

SALBP-F is considered as a feasibility problem for a given combination of cycle time and the number of stations. SALBP-I and SALBP-II models have a dual relationship while the first tries to minimize the number of stations with respect to a given cycle time, the second, tries to minimize the cycle time with regard to a given number of stations. Additionally, SALB-E model aims to simultaneously minimize the number of stations as well as the cycle time so that the efficiency has to be maximized.

The assembly line can be also organized into other classifications depending on its layout as well as variety of products. In this direction, in layout of production line perspective, it can be classified into two groups: straight and U-shaped assembly lines, while with regard to number of different products it might be categorized into the single model (one product) and mixed model (multi-product) assembly lines.

* Corresponding author. Tel.: +98 9112116291; fax: +98 35257757.
E-mail address: mostafa.alavidoost@aut.ac.ir (M.H. Alavidoost).

http://dx.doi.org/10.1016/j.asoc.2015.11.025
1568-4946/

Table 1
Versions of SALBP [4].

		Cycle time	
		Given	Minimize
The number of stations	Given	SALBP-F	SALBP-II
	Minimize	SALBP-I	SALBP-E

The straight assembly line has been considered as the most important part of traditional mass production until introducing U-shaped assembly line as a result of continuous improvement and cost reduction proposed in just-in-time (JIT) system [5]. There are several advantages associated with U-shaped assembly line in comparison with the straight one, such as flexibility increase, productivity increase, quality improvement, and lower WIP inventory.

The main feature which makes the U-shaped assembly line different from a straight one is that in the former, the entrance and exit are located in the same position.

The two main problems investigated in this paper consist of the simple assembly line balancing problem (SALB) and U-shaped version of SALB (SULB) such that SALB and SULB refer to straight and single model assembly lines, and U-shaped and single model assembly lines problems, respectively.

The remainder of this paper proceeds as follows: Section 2 is devoted to report the relevant literature in this area. In Section 3, proposed fuzzy MOLP models for the SALB and SULB are developed. A novel interactive fuzzy approach as a solution procedure for solving the problem is provided in Section 4. In Section 5, the validity of the proposed model and the solution method are evaluated through illustrative examples. Finally, conclusion remarks and proposal for future works are presented in Section 6.

2. Literature review

The SALB problem was first formulated by Salveson [6]. After that, considerable researches have been spent to address this problem and they have been reviewed in [4,7–11]. Thus, among ample of works dedicated to this context, some of the most recent ones are provided as follows.

Ağpak and Zolfaghari [12] presented a mixed integer programming model for the parallel two-sided ALBP. Gupta [13] optimized a multi model assembly line. Moreira et al. [14] proposed a new assembly line balancing problem namely assembly line worker integration and balancing problem (ALWIBP). Roshani and Giglio [15] presented a mathematical programming formulation for cost-oriented multi-manned ALBP. Triki et al. [16] presented an ALBP in a real-world automotive cables manufacturer company. Choudhary and Agrawal [17] improved the performance measures of mixed model assembly line. Pereira [18] introduces new lower bounds for the simple ALBP and analyzed their performance.

Compared with SALB literature, SULB has been relatively less taken into account in the literature. The first dynamic programming formulation for the SULB problem was proposed by Miltenburg and Wijngaard [19]. They also presented a modified heuristic approach to solve the problem. The integer programming (IP) formulation for SULB problem was first developed by Urban [20]. The first multi-criteria approach to SULB models was presented by Gökçen and Ağpak [21]. In their research, a goal programming approach is presented so that it was a combination of IP formulation of Urban [20] and goal programming model proposed in Deckro and Rangachari [22].

Fattahi and Turkay [23] presented a revised formulation for SULB model and tested their revision formulation on same benchmark problems. Fattahi et al. [24] presented a novel integer programming formulation for SULB and enhanced the efficiency of the LP relaxation of their new formulation.

2.1. Fuzzy models of ALBP

In many cases of the real world, we have to make decision with respect to vague or uncertain conditions. This uncertainty and vagueness can be resulted from either goals values or imprecision in problem's data. In assembly line problem, the uncertainty associated with real data is resulted from both human and machine factors which can be estimated only using uncertain values [25]. In this regard, Fuzzy Set Theory is considered as one of the most promising approaches to deal with uncertainty [26]. In other words, fuzzy objectives can be employed to satisfy the aspiration level of imprecise goals. Also, fuzzy numbers can appropriately reflect the vagueness corresponding to real data.

In comparison with crisp ALBP, studying the ALBP in uncertain environment has received less attention in the literature so that there are few studies addressing the assembly line problem in uncertain environment [4,10].

Tsujimura et al. [27] and Gen et al. [28] were the first who developed a genetic algorithm as a solution approach to address fuzzy SALB (*f*-SALB) problem with considering fuzzy task processing times. Other works provide solution approach for both *f*-SALB and *f*-SULB problems such as [25,29–36]. Also, some researchers applied stochastic approach to deal with uncertainty in the modeling process. In this line some most recent studies in this context are presented in [37–39].

Having employed fuzzy programming, La Scalia [40] conducted the first and the last study in which the *f*-SALB problem was formulated and solved following an exact solution procedure where task processing times were considered as fuzzy numbers. Zhang and Cheng's [41] is the only work dealt with the *f*-SULB problem considering fuzzy parameters.

Applying fuzzy goal programming (FGP) method, Toklu and özcan [42] developed a FGP approach to consider the SULB problem. Use of binary FGP (BFGP) approach Kara et al. [43] proposed a mathematical formulation as well as a solution approach to deal with both the multi-objective SALB and SULB problems. Hazır and Dolgui [44] employed a robust formulation for U-type assembly line balancing with interval data.

As mentioned above, due to the importance of SALB problem, many works have been dedicated to this area of research. However, despite a significant advantage of SULB to increase the efficiency of assembly line compared to SALB, the SULB received less effort in the literature so far. This issue reveals a considerable gap exists in the literature where the presented study attempts to fill that. Additionally, due to importance of fuzzy approach in modeling the real world problems and lack of a comprehensive model for the SALB and SULB problems wherein fuzzy parameters and fuzzy goals are considered at the same time, this paper intended to response the need of comprehensive model for *f*-SALB and *f*-SULB in which fuzzy parameters as well as fuzzy goals are taken into account simultaneously.

One of the methods employed for modeling the fuzzy goals is to apply fuzzy interactive approach that because of its capability to directly communicate with DM has been received considerable attention by recent researchers. In fact, this method enables the DM with the least possible information of the problem domain to achieve preferred compromise solution so that this issue led the presented study to utilize the fuzzy interactive approach in the modeling process.

To the best of our knowledge, there is no work in the literature dedicated to model multi-objective SALB and SULB problems regarding fuzzy interactive approach. Therefore, this paper constitutes a first framework to deal with assembly line balancing problem when uncertain condition is held. In this study, after developing a new BOFMILP model for SALB and SULB problems, a novel interactive fuzzy approach is proposed as a solution.

3. Problem formulation

This section provides mathematical formulation for both *f*-SALB and *f*-SULB problems.

3.1. Notation

The following indices, parameters, decision variables, and indicator variables are used in the proposed MIP formulation:

Indices:

i, a, b	Task
j	Station
k	Objective function

Parameters:

I	The set of tasks, $I = \{i \mid i = 1, 2, \ldots, n\}$
J	The set of station, $J = \{j \mid j = 1, 2, \ldots, m_{\max}\}$
P	The set of precedence relations
$\tilde{t}_i$	Fuzzy task processing time i
θ_k	Importance and weight of kth objective
γ	Coefficient of compensation

Decision variables:

nWS	Number of utilized station
$\tilde{c}$	Fuzzy cycle time of assembly line
x_{ij}	1, **if** ith task of the original diagram is assigned to station j, **otherwise** 0.
y_{ij}	1, **if** ith task of the phantom diagram is assigned to station j, **otherwise** 0.
X_j	1, **if** at least one task is assigned to station j, **otherwise** 0.
f_k	kth objective function
$\mu_k(x)$	kth linear membership function (MF)
λ_0	Minimum level of satisfaction
M^∞	A very large positive number

3.2. Formulation of f-SALB problem

A mathematical formulation for SALB problem is presented in this section.

3.2.1. Objective functions

Generally, the assembly line balancing problem comprises various conflicting objective functions so that they have to be optimized simultaneously. In regard to this statement, among numerous objective functions proposed in the literature for ALBP [21,25], this paper objects to optimize the following objective functions:

The first one is to minimize the number of stations as can be formulated in equation

$$f_1 = \min(nWS) = \min\left\{\sum_{j=1}^{m_{\max}} X_j\right\} \tag{1}$$

whereas the second one concerns to minimize the fuzzy cycle time according to Eq. (2).

$$\tilde{f}_2 = \min(\tilde{c}) \tag{2}$$

3.2.2. Constraints

In the presented study, a combination of constraints suggested by [21,22,28] as well as those specified for this problem is used as model constraints. In particular, new set of constraints considering fuzzy coefficients and variables can be formulated as follows:

$$\sum_{j=1}^{m_{\max}} x_{ij} = 1, \quad \forall i \in I \tag{3}$$

$$\sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)x_{aj} - \sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)x_{bj} \geq 0, \quad \forall (a, b) \in P \tag{4}$$

$$\sum_{i=1}^{n} \tilde{t}_i(x_{ij}) \leq \tilde{c}, \quad \forall j \in J \tag{5}$$

$$\sum_{i=1}^{n} \tilde{t}_i(x_{ij}) + (1 - U_j)M^\infty \geq \tilde{c}, \quad \forall j \in J \tag{6}$$

$$\sum_{j=1}^{m_{\max}} U_j = 1 \tag{7}$$

$$\sum_{i=1}^{n} (x_{ij}) - n * X_j \leq 0, \quad \forall j \in J \tag{8}$$

$$\tilde{c} \text{ is a TFN}, \quad x_{ij}, X_j, U_j \in \{0, 1\}, \quad \forall j \in J, \quad \forall i \in I \tag{9}$$

Constraint (3) is called assignment constraint in which ensures that all tasks will be allocated to at least one corresponding station. Meanwhile it guarantees that each task will be allocated only to one the station. Constraint (4) is related to the precedence constraints. According to [28], the maximum fuzzy processing time required for stations is defined as a fuzzy cycle time of the assembly line ($\tilde{c} = \max_j\{\sum_{i=1}^{n} \tilde{t}_i(x_{ij})\}$) and constraints (5)–(7) are employed to convert this constraint into the linear form. Also, Constraint (8) demonstrates that station constraint. Finally, Constraint (9) represents the feasible region associated with the variables.

3.3. Formulation of f-SULB problem

A mathematical formulation for SULB problem is presented in this section.

3.3.1. Objective functions

The objective functions considered for the SULB problem are defined similar to those in the SALB one (Eqs. (1) and (2)).

3.3.2. Constraints

In this paper, the constraints suggested by Gökçen and Ağpak [21] are considered as model constraints. So, new set of constraints regarding fuzzy coefficients and new variables can be formulated as follows:

$$\sum_{j=1}^{m_{\max}} (x_{ij} + y_{ij}) = 1, \quad \forall i \in I \tag{10}$$

$$\sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)x_{aj} - \sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)x_{bj} \geq 0, \quad \forall (a, b) \in P \tag{11}$$

$$\sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)y_{bj} - \sum_{j=1}^{m_{\max}} (m_{\max} - j + 1)y_{aj} \geq 0, \quad \forall (a, b) \in P \tag{12}$$

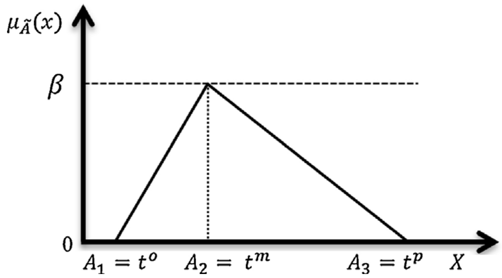

Fig. 1. Membership function of TFN.

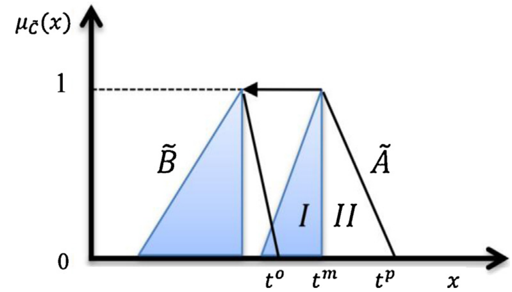

Fig. 2. The minimization strategy for a fuzzy objective.

$$\sum_{i=1}^{n} \tilde{t}_i(x_{ij}+y_{ij}) \le \tilde{c}, \quad \forall j \in J \tag{13}$$

$$\sum_{i=1}^{n} \tilde{t}_i(x_{ij}+y_{ij}) + (1-U_j)M^{\infty} \ge \tilde{c}, \quad \forall j \in J \tag{14}$$

$$\sum_{j=1}^{m_{\max}} U_j = 1 \tag{15}$$

$$\sum_{i=1}^{n} (x_{ij}+y_{ij}) - n * X_j \le 0, \quad \forall j \in J \tag{16}$$

$$\tilde{c} \text{ is a TFN}, \quad x_{ij}, y_{ij}, X_j, U_j \in \{0, 1\}, \quad \forall j \in J, \quad \forall i \in I \tag{17}$$

Constraint (10) is also called assignment constraint. Constraints (11) and (12) satisfy the precedence relations among tasks and ensure that only the tasks which all predecessor or all successors are assigned to an earlier or the current station can be assigned to the current station. Constraints (13)–(15) are equivalent to the corresponding ones in the SALB problem (constraints (5)–(7)). Constraint (16) represents the station constraint and Eq. (17) represents the feasible area related to the variables.

4. Solution approach

In this study, a two-phase approach is adopted to solve fuzzy ALBP. While in the first phase, the original problem is converted into an equivalent auxiliary crisp multi-objective mixed integer linear programming (MOMILP) model, in the second phase, a novel two-phase interactive fuzzy programming approach is proposed to achieve a satisfactory compromise solution.

4.1. Phase I: an auxiliary crisp MOMILP model

In this work, TFNs are employed in order to account uncertainty associated with the task processing times. Generally, a triangular fuzzy processing time is defined by a triplet points $\tilde{t} = (t^o, t^m, t^p)$. A membership function (MF) for TFN and its corresponding equation are presented in Fig. 1 and Eq. (18), respectively [45,46]. It is worth noting that we choose the TFN due to its flexibility of the fuzzy arithmetic operations, computational efficiency and simplicity in data acquisition.

$$\mu(x) = \begin{cases} \beta \dfrac{x - A_1}{A_2 - A_1}, & A_1 \le x \le A_2 \\ \beta \dfrac{A_3 - x}{A_3 - A_2}, & A_2 \le x \le A_3 \\ 0 & O.W. \end{cases}, \quad 0 < \beta \le 1 \tag{18}$$

As can be seen from Fig. 1, a fuzzy triangular MF consists of three main elements including:

- The most optimistic value (c^o)
- The most possible value (c^m)
- The most pessimistic value (c^p)

As clear from Fig. 1, the two first ones are located in the upper bound and lower bound of feasible area respectively.

While the most optimistic and pessimistic values are located at the beginning and end of the interval, respectively, they have the lowest membership degree ($\mu_{\tilde{t}}(t^p) = \mu_{\tilde{t}}(t^o) = 0$). The most possible value is posited within the interval and has the maximum degree of membership ($\mu_{\tilde{t}}(t^m) = \beta$). It is assumed that all other points located between these three points have membership degree whose values vary linearly between [0, β] (It should be noted that if the MF is normal, the value for β will be set as 1).

4.1.1. An approach to deal with fuzzy objective function

The second objective function is defined by a TFN ($\tilde{f}_2 = \min\{\tilde{c}\} = \min\{(c^o, c^m, c^P)\}$). There have been several methods proposed in the literature to deal with optimizing fuzzy objective functions in which they are represented by fuzzy numbers. According to method suggested by Lai and Hwang [47], in order to solve a fuzzy integer programming problem, an auxiliary Multi-Objective Linear Programming (MOLP) approach with converting a fuzzy objective into three independent crisp ones is applied. According to this strategy, this study intends to simultaneously minimize the most possible value of the imprecise processing time $\{f^1 = \min(c^m)\}$, maximize the possibility of obtaining lower processing time $\{f^2 = \max(c^m - c^o)\}$, and minimize the risk of obtaining higher objective value $\{f^3 = \min(c^p - c^m)\}$. Fig. 2 depicts the minimization strategy for an imprecise objective [48]. As one can see from Fig. 2, compared with A, the objective function B is much preferable as it simultaneously meets the requirements of all three objectives.

Therefore, the objective functions associated with the Auxiliary MOLP are defined by Eqs. (19)–(22).

$$f_1 = \min(nWS) = \min \left\{ \sum_{j=1}^{m_{\max}} X_j \right\} \tag{19}$$

$$f_2 = \min(c^m) \tag{20}$$

$$f_3 = \max(c^m - c^o) \tag{21}$$

$$f_4 = \min(c^p - c^m) \tag{22}$$

4.1.2. An approach to deal with fuzzy constraints

4.1.2.1. Treating fuzzy constraints in SALBP. In this paper, we adopt the procedure suggested by Wang and Liang [48]. In this regard, when both right and left sides of constraint are fuzzy numbers, three equivalent constraints are developed.

Use of this method, constraints (5) and (6) are converted to the set of constraints (23)–(25) and (26)–(28) respectively. Doing so, the following equations are obtained:

$$\sum_{i=1}^{n} t_i^p(x_{ij}) \le c^p, \quad \forall j \in J \tag{23}$$

$$\sum_{i=1}^{n} t_i^m(x_{ij}) \le c^m, \quad \forall j \in J \tag{24}$$

$$\sum_{i=1}^{n} t_i^o(x_{ij}) \le c^o, \quad \forall j \in J \tag{25}$$

$$\sum_{i=1}^{n} t_i^p(x_{ij}) + (1-U_j^1)M^{\infty} \ge c^p, \quad \forall j \in J \tag{26}$$

$$\sum_{i=1}^{n} t_i^m(x_{ij}) + (1-U_j^2)M^{\infty} \ge c^m, \quad \forall j \in J \tag{27}$$

$$\sum_{i=1}^{n} t_i^o(x_{ij}) + (1-U_j^3)M^{\infty} \ge c^o, \quad \forall j \in J \tag{28}$$

Also, constraint (7) is converted to constraint (29) as presented below:

$$\sum_{j=1}^{m_{\max}} U_j^l = 1, \quad l=1,2,3 \tag{29}$$

Consequently, having transformed our *f*-SALB model into an auxiliary crisp MOMILP one, Eqs. (19)–(22) as a set of objectives as well as Eqs. (3), (4), (8), (9), and (23)–(29) as a set of constraints are obtained.

4.1.2.2. Treating fuzzy constraints in SULBP. Recalling the procedure mentioned before, constraints (13) and (14) are converted to the set of constraints which can be found in Eqs. (30)–(32) and Eqs. (33)–(35) respectively, and also constraint (15) is converted to constraint (36).

$$\sum_{i=1}^{n} t_i^p(x_{ij}+y_{ij}) \le c^p, \quad \forall j \in J \tag{30}$$

$$\sum_{i=1}^{n} t_i^m(x_{ij}+y_{ij}) \le c^m, \quad \forall j \in J \tag{31}$$

$$\sum_{i=1}^{n} t_i^o(x_{ij}+y_{ij}) \le c^o, \quad \forall j \in J \tag{32}$$

$$\sum_{i=1}^{n} t_i^p(x_{ij}+y_{ij}) + (1-U_j^1)\cdot M^{\infty} \ge c^p, \quad \forall j \in J \tag{33}$$

$$\sum_{i=1}^{n} t_i^m(x_{ij}+y_{ij}) + (1-U_j^2)\cdot M^{\infty} \ge c^m, \quad \forall j \in J \tag{34}$$

$$\sum_{i=1}^{n} t_i^o(x_{ij}+y_{ij}) + (1-U_j^3)\cdot M^{\infty} \ge c^o, \quad \forall j \in J \tag{35}$$

$$\sum_{j=1}^{m_{\max}} U_j^l = 1, \quad l=1,2,3 \tag{36}$$

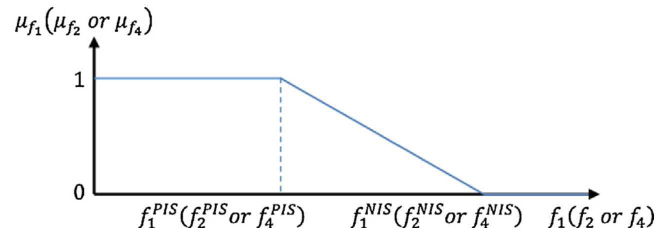

Fig. 3. Linear membership functions of f_1, f_2 and f_3.

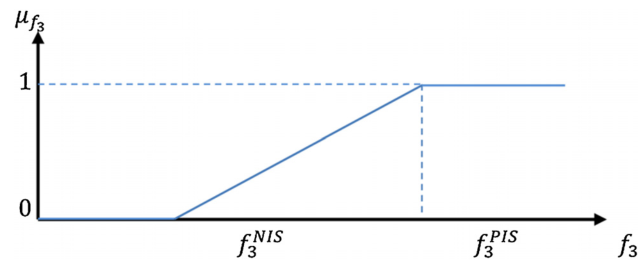

Fig. 4. Linear membership function of f_3.

Finally, *f*-SULB is transformed into an auxiliary crisp MOMILP one, involving Eqs. (19)–(22) as set of objectives as well as Eqs. (10)–(12), (16),(17), and (30)–(36), as set of constraints.

4.2. Phase II: interactive fuzzy programming approach

4.2.1. Fuzzy programming

In this paper, an efficient interactive fuzzy programming approach is used to achieve an efficient compromise solution. Different methodologies are developed for solving multi-objective optimization problems such as the weighted-sum method, the ε-constraint method, the goal-programming method and fuzzy method [49]. Fuzzy decision making and Fuzzy Programming are two methods that have been proposed by Bellman and Zadeh [50] and Zimmermann [51] respectively, to solve MOLP in which both methods tend to obtain the positive ideal solutions (*PIS*) and the negative ideal solutions (*NIS*) of all the corresponding objective functions. These values are determined either by the DM's preferences (between the worst and best values of each function) or are equal to the worst and best values of each function (Eq. (37)).

$$\begin{aligned} &f_1^{PIS} = \min f_1 = \min(nWS), && f_1^{NIS} = \max f_1 = \max(nWS) \\ &f_2^{PIS} = \min f_2 = \min(c^m), && f_2^{NIS} = \max f_2 = \max(c^m) \\ &f_3^{PIS} = \max f_3 = \max(c^m - c^o), && f_3^{NIS} = \min f_3 = \min(c^m - c^o) \\ &f_4^{PIS} = \min f_4 = \min(c^P - c^m), && f_4^{NIS} = \max f_4 = \max(c^P - c^m) \end{aligned} \tag{37}$$

Each objective function corresponds to an equivalent linear MF which can be obtained using Eqs. (38)–(41) and the graphs for these MFs are depicted in Figs. 3 and 4.

$$\mu_1(x) = \begin{cases} 1 & \text{if} \quad f_1 \le f_1^{PIS} \\ \dfrac{f_1^{NIS} - f_1}{f_1^{NIS} - f_1^{PIS}} & \text{if} \quad f_1^{PIS} \le f_1 \le f_1^{NIS} \\ 0 & \text{if} \quad f_1 \ge f_1^{NIS} \end{cases} \tag{38}$$

$$\mu_2(x) = \begin{cases} 1 & \text{if} \quad f_2 \le f_2^{PIS} \\ \dfrac{f_2^{NIS} - f_2}{f_2^{NIS} - f_2^{PIS}} & \text{if} \quad f_2^{PIS} \le f_2 \le f_2^{NIS} \\ 0 & \text{if} \quad f_2 \ge f_2^{NIS} \end{cases} \tag{39}$$

$$\mu_3(x) = \begin{cases} 1 & if \quad f_3 \le f_3^{PIS} \\ \dfrac{f_3^{NIS} - f_3}{f_3^{NIS} - f_3^{PIS}} & if \quad f_3^{PIS} \le f_3 \le f_3^{NIS} \\ 0 & if \quad f_3 \ge f_3^{NIS} \end{cases} \quad (40)$$

$$\mu_4(x) = \begin{cases} 1 & if \quad f_4 \le f_4^{PIS} \\ \dfrac{f_4^{NIS} - f_4}{f_4^{NIS} - f_4^{PIS}} & if \quad f_4^{PIS} \le f_4 \le f_4^{NIS} \\ 0 & if \quad f_4 \ge f_4^{NIS} \end{cases} \quad (41)$$

where in Eqs. (38)–(41) $\mu_k(x)$ represents the linear MF for the kth fuzzy objective.

Using max–min operator, Zimmermann [51] solved the MOLP model but the solution of this procedure might be neither unique nor efficient [52,53]. To eliminate this deficiency, some researches have been conducted in the literature. Among them, Lai and Hwang [52] proposed an augmented max–min approach (LH approach) and Li et al. [53] suggested two-phase fuzzy method (LZL approach) in order to remove such drawback.

Since in the last two methods, great amount of attention was paid to increase the minimum aspiration level of goals in comparison to the objective weights, they usually lead to produce solutions which are not compatible for the DM who prefers to prioritize the objectives. In other words, in these methods the effect of objective weight did not receive significant attention.

In order to alleviate the deficiency related to prioritize the goals, Selim and Ozkarahan [54] presented a modified version of Werners' approach [55] (MW approach) and Torabi and Hassini [56] (TH approach) proposed a combination of the MW and LH methods in which the coefficient of compensation was offered to account the effect of change in objective weights. In light of these two approaches, since a small change in the coefficient of compensation may yield a large amount of variability in the obtained solution (especially for its lower values, smaller than 0.3), it is difficult to set this value accurately. In addition to these works, Demirli and Yimer [57] provided a framework to consider prioritizing of objectives, but since it utilized the max–min operator, it might result to the solutions that are neither unique nor efficient. A brief overview of these methods has been provided in Appendix A.

Consequently, this study tried to develop a new single-phase approach removing above deficiencies that led to a novel fuzzy approach as well as a new model formulation to solve the ALBP models. In summary, the proposed interactive solution procedure to solve the original MOFMILP model is provided in the next section.

4.2.2. The proposed interactive fuzzy programming approach

The significant advantage of interactive approaches is related to their ability to provide a systematic framework enabling the DM to interactively adjust the parameters according to his/her preferences until an efficient satisfactory solution is reached. In other words, the DM adjusts interactively the search direction during the solution procedure, until the efficient solution satisfies the DM's preferences [58].

The proposed fuzzy programming approach extends the method developed by Abd El-Wahed and Lee [58]. This approach consists of ten steps which includes the following steps:

Step 1: Specify an appropriate the pattern of triangular possibility distribution for all uncertain data and construct the fuzzy MOLP problem.

Step 2: Transform the fuzzy objectives in to crisp ones according to procedure described in 4-1-1.

Step 3: Convert the Fuzzy constraints in to crisp ones according to procedure explained in 4-1-2.

Step 4: Perform the optimization process for each objective function. If the solutions obtained from this process are equal, then select one of them and deliver to the DM and go to step 10. Otherwise, go ahead and continue the procedure on step 5.

Step 5: Considering the obtained solutions resulted from previous step, having the *PIS* value associated with each objective, determine the negative ideal solution (*NIS*) for each objective function by solving the corresponding MILP model according to Eq. (37).

Step 6: Specify the linear MF correspond to each objective function according to Eqs. (38)–(41).

Step 7: Transform the MOMILP problem into an equivalent MILP by means of the following proposed auxiliary crisp formulation (Eq. (42)) and solve it (It is worth noting that the proposed auxiliary crisp formulation is a hybrid of the MW, LH and DY methods that will be discussed in next section).

$$\begin{aligned} &\max \lambda(x) = \lambda_0 + \delta \sum_{k=1}^{K} \theta_k \lambda_k \\ &S.t: \theta_k \lambda_0 + \lambda_k \le \mu_k(x), \quad k = 1, \ldots, K \\ &x \in F(x) \end{aligned} \quad (42)$$

where, $\mu_k(x)$ refers to the satisfaction level of objective and δ denotes to small positive number which is normally set to 0.01. Additionally, the relative importance of kth objective is represented by θ_k which is determined according to DM's preference so that $\sum_k \theta_k = 1, \theta_k > 0$. Moreover, in the above formulation, the compromise degree among the objectives and the minimum satisfaction level of objectives is controlled by λ_0. x indicates decision variables and $F(x)$ should be interpreted as feasible region which covers all the constraints.

Step 8: Deliver the obtained solution to the DM; if the solution meets the preference of the DM, then it is considered as compromise solutions and go to step 10. Otherwise, go to step 9.

Step 9: The model should be interactively revised according to DM's preference to reach the desirable solution. This process repeats until the solution approach lead to produce the satisfactory solution. In this regard, the changes, which are permitted, are as follows:

- For maximization objectives, increase the *NIS*.
- For minimization objectives, decrease the *PIS*.
- Changes in weights of objective

It should be noted that, since any changes in the *NIS* value can lead to the large variation in the obtained result, the size of change in the *NIS* should be as small as possible to prevent from locating in infeasible region.

Step 10: Stop, the algorithm has reached to the compromise solution.

A flowchart for the proposed interactive fuzzy programming approach is displayed in Fig. 5.

5. Numerical illustration

To test and assess the validity and efficiency of the proposed model, a computational study was performed. In this line, first a test problem suggested in [43] is considered. Then, the SALB and SULB problems are solved by means of the proposed interactive fuzzy approach. After that, the obtained solutions are evaluated over those in the test problem. Finally, a comparison study is conducted between our proposed fuzzy approach and other approaches in open literature. The properties of the test problem are summarized

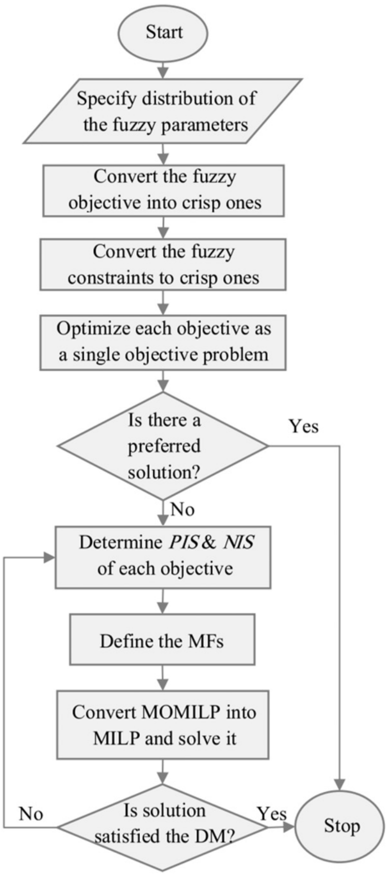

Fig. 5. Flowchart of proposed interactive fuzzy programming approach.

Table 2
Fuzzy processing time of tasks for test problem 1.

Task no.	Processing time	Task no.	Processing time
1	(4, 5, 6)	5	(3, 4, 5)
2	(3, 4, 6)	6	(2, 3, 4)
3	(6, 7, 8)	7	(1, 2, 3)
4	(2, 3, 4)	8	(2, 3, 4)

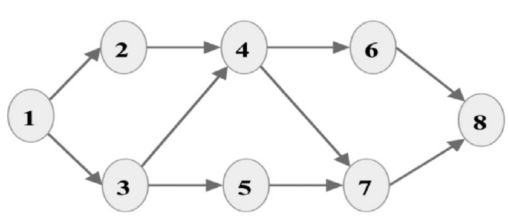

Fig. 6. Precedence diagram of test problem 1.

in Table 2 whilst Fig. 6 displays its precedence diagram (note that, we convert crisp processing times of test problem into TFNs).

5.1. Proposed solution approach for solving the problem

The interactive solution method using the proposed two-phase approach is described as follows:

Step 1: In this study, the triangular MF is adopted to represent the imprecise task processing times (Table 2). Mathematical formulations of the f-SALB and f-SULB problems are presented in Sections 3.2 and 3.3 respectively.

Step 2: Convert the original Fuzzy objective function ($\tilde{f}_2 = \min(\tilde{c})$) in to three independent crisp objective functions as in Eqs. (20)–(22).

Table 3
First optimization points for considered problems.

Problem	*WS*	Assigned tasks	Fuzzy cycle time of each station	$\tilde{c}$
First solution for SALB	1	1, 2	(7,9,12)	(9,11,15)
	2	3, 5	(9,11,13)	
	3	4, 6, 7, 8	(7,11,15)	
First solution for SULB	1	1, 6, 8	(8,11,14)	(8,11,14)
	2	2, 5, 7	(7,10,14)	
	3	3, 4	(8,10,12)	

Step 3: Transform the fuzzy constraints into the corresponding crisp ones (Section 4.1.2), and formulate the auxiliary crisp MOMILP model.

Step 4: The optimization process was performed separately for each objective, but no unique solution is achieved. Thus, the optimized objectives are provided as:

$$\min(nWS) = 1, \quad \min(c^m) = 7,$$
$$\max(c^m - c^o) = 8, \quad \min(c^P - c^m) = 1$$

Step 5: Determine the *PIS* and *NIS* for each objective function. These values are given as follows:

$$\begin{aligned}
&f_1^{PIS} = \min(nWS) = 1, && f_1^{NIS} = \max(nWS) = 8\\
&f_2^{PIS} = \min(c^m) = 7, && f_2^{NIS} = \min(c^m) = 31\\
&f_3^{PIS} = \max(c^m - c^o) = 8, && f_3^{NIS} = \min(c^m - c^o) = 1\\
&f_4^{PIS} = \min(c^P - c^m) = 1, && f_4^{NIS} = \max(c^P - c^m) = 9
\end{aligned}$$

Step 6: Using the obtained values of the *PIS* and *NIS* for each objective function, the corresponding Linear MF can be specified as presented in Eqs. (38)–(41).

Step 7: Recalling Eq. (42), transform the MOMILP model to the corresponding MILP one.

Unlike some studies in the literature, assuming all objective functions have the same relative importance in DM's point of view, in this study, to make it better suited for practical applications, the different relative importance for objective weights are considered. To so do, it is assumed that the preference information corresponding to the relative importance of the objective functions is specified by the decision maker as: $\theta_i = (0.45, 0.45, 0.05, 0.05)$. Indeed, having utilized the DM's preferences, the concept of the most possible values proposed by Lai and Hwang [59] is applied in order to determine the weights of $f_2, f_3, \& f_4$. The reason for that lies in the fact that the most possible value is generally the most important one, so this value should receive greater weight [48].

Both SALB and SULB problems are implemented in LINGO 11 software. The obtained solutions are presented in Table 3.

Step 8: Let's suppose that from the DM's point of view, the obtained solution is not the satisfactory one due to its large cycle time.

Step 9: Consider the DM has intended to adjust the cycle time according to his/her preference. To this end, he/she may decrease the *NIS* value to the second objective function (i.e. minimizing the most likely value) and after some iteration it is set to: 10.5.

Consequently, the final solutions produced by the proposed algorithm are demonstrated in Table 4. In addition, the optimum layouts for the straight and U-shaped assembly lines, offered by the proposed approach, are depicted in Fig. 7.

Step 10: Stop, the algorithm has reached to the compromised solution for ALBP.

According to the several studies devoted to context of U-shaped assembly line, it is found that when all condition are the same in assembly line problem, the number of station required for straight line is greater or equal to the number of station required for the

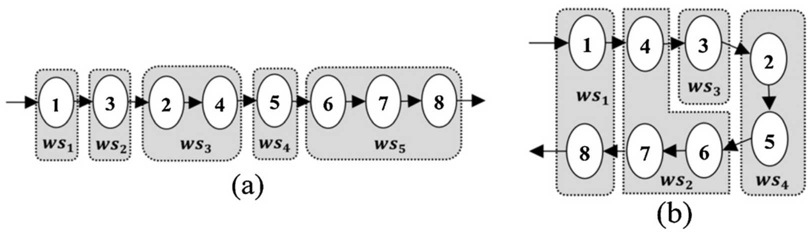

Fig. 7. Optimum layout for Straight assembly line (a), optimum layout for U-shaped assembly line (b).

U-shaped one [43]. The results obtained from this study also support this statement.

5.2. Performance analysis of the proposed fuzzy approach

In order to assess the performance of the presented fuzzy approach in which a MOLP is converted to the single objective linear problem, comparative study considering five well-known approaches in the literature (i.e. the LZL, LH, MW, TH and DY) is conducted in term of distance measure (43).

$$D_p(\theta, K) = \left[\sum_{k=1}^{K} \theta_k{}^p (1-\mu_k(x))^p\right]^{1/p}, \quad p \geq 1 \text{ and integer} \tag{43}$$

where the power p indicates a distance parameter and its most operationally important values are: 1, 2, & ∞ (Eqs. (44)–(46)).

$$D_1(\theta, K) = 1 - \sum_{k=1}^{K} \theta_k \mu_k(x) \tag{44}$$

$$D_2(\theta, K) = \left[\sum_{k=1}^{K} \theta_k{}^2 (1-\mu_k(x))^2\right]^{1/2} \tag{45}$$

$$D_\infty(\theta, K) = \max_k \{\theta_k (1-\mu_k(x))\} \tag{46}$$

While the Manhattan distance (D_1) as well as the Euclidean distance (D_2) are geometrically considered as the longest and shortest distances respectively and in numerical sense, the Tchebycheff distance (D_∞) is the shortest one. In fact, D_1 is the weighted sum of distance from goal (i.e. distance from value 1) and whatever μ_ks are larger, it takes less value. This rule is similarly holds for D_2, where D_2 not only checks the distance from goal, but also evaluates the equality value of μ_ks. In other words, if value is equal in two different solutions, for solution that the μ_ks value is more close together, D_2 takes less value (of course, in the same value of μ_ks). Indeed, at formula (43), the more increase in p value, the more penalty is assigned to smaller μ_k and this leads μ_ks to take more closed value. In other words, if $p > 1$, this means equality in satisfaction of the goals is considered as well. To make clear this difference, an example is presented in Table 5. As one can see, although D_1 has the same value in the two solutions but for more balance of solution 1, D_2 and D_∞ take less value for that.

In the presented paper, in order to analyze the performance of considered methods (i.e. The LZL, LH, MW, TH and DY) three test problems have been considered where in the following, the selected approaches will be examined over them.

5.2.1. Test problem 1

This test problem is the same as presented in [43] (Table 2 and Fig. 6). In this work, to evaluate the effect of change in the weight of objectives on the performance of the five aforementioned methods, four different weights are taken in to account in our implementation. Table 6 summaries the result produced by these methods.

As seen in Table 6, in the both methods LZL and LH, the aspiration level of the objectives is not affected by change of the weights. In other words, these methods are not sensitive to the objective weights, which is not desirable for the DM. In addition, the LZL and LH methods showed lower performance in comparison to three other ones with respect to distance measure. As stated earlier, the obtained solutions using the DY approach are neither efficient nor unique. This issue can be confirmed by comparison between the DY method and our proposed approach as provided in Table 6. Among the other three approaches (i.e. the MW, TH and the proposed approach), in many cases, the TH method provides the solutions having higher distance in comparison with two other ones. Moreover, it can be found that there is no significant difference between the solutions found by the MW and proposed approaches with respect to distance measures. In addition, while in some cases the proposed approach outperforms the MW, this manner is not the true in all conditions. Of course, it should be noted that in the many cases, D_2 and D_∞ take less value for the proposed approach in comparison with MW method which indicates a better balance in satisfaction of the objectives.

Further investigations related to the performance of the fuzzy methods in terms of distance measure as well as satisfaction level of the objective functions along with different weights are provided in Appendix B-Test Problem 1 (Figs. 10–13 and 14–17, respectively).

5.2.2. Test problem 2

This test problem has presented in paper [27]. The task processing times for this problem and also the precedence constraints are represented in Table 7 and Fig. 8, respectively.

Table 4
Final solution for considered problems.

Problem	WS	Assigned tasks	Fuzzy cycle time of each station	$\tilde{c}$
Final solution for SALB	1	1	(4,5,6)	(6,8,11)
	2	3	(6,7,8)	
	3	2, 4	(5,7,10)	
	4	5	(3,4,5)	
	5	6, 7, 8	(5,8,11)	
Final solution for SULB	1	1, 8	(6,8,10)	(6,8,11)
	2	4, 6, 7	(5,8,11)	
	3	3	(6,7,8)	
	4	2, 5	(6,8,11)	

Table 5
An example for difference of distance measures.

	θ	Solution 1	Solution 2
μ_1	0.25	0.75	0.5
μ_2	0.25	0.75	0.9
μ_3	0.25	0.75	0.65
μ_4	0.25	0.75	0.95
D_1		0.25	0.25
D_2		0.016	0.024
D_∞		0.004	0.016

Table 6
Comparison between fuzzy approaches on test problem 1 (δ = 0.01, γ = 0.4).

Weights	Approaches	μ_k				D_1	D_2	D_∞
		μ_1	μ_2	μ_3	μ_4			
$\theta 1$ = (0.45, 0.45, 0.05, 0.05)	LH	0.71	0.67	0.57	0.5	0.33	0.20	0.15
	LZL	0.71	0.67	0.57	0.5	0.33	0.20	0.15
	MW	0.71	0.83	0.14	0.63	0.27	0.16	0.13
	TH	0.71	0.79	0.43	0.63	0.27	0.16	0.13
	DY	0.71	0.75	0.29	0.5	0.30	0.18	0.13
	Proposed approach	0.71	0.83	0.14	0.63	0.27	0.16	0.13
$\theta 2$ = (0.2, 0.6, 0.05, 0.15)	LH	0.71	0.67	0.57	0.5	0.35	0.22	0.20
	LZL	0.71	0.67	0.57	0.5	0.35	0.22	0.20
	MW	0.43	1	0	0.75	0.20	0.13	0.11
	TH	0.71	0.79	0.43	0.63	0.27	0.15	0.13
	DY	0.43	0.96	0.14	0.75	0.22	0.13	0.11
	Proposed approach	0.43	0.96	0.14	0.75	0.22	0.13	0.11
$\theta 3$ = (0.24, 0.32, 0.18, 0.26)	LH	0.71	0.67	0.57	0.5	0.38	0.20	0.13
	LZL	0.71	0.67	0.57	0.5	0.38	0.20	0.13
	MW	0.71	0.79	0.14	0.88	0.32	0.18	0.15
	TH	0.71	0.67	0.57	0.5	0.38	0.20	0.13
	DY	0.57	0.79	0.43	0.63	0.37	0.19	0.10
	Proposed approach	0.71	0.79	0.43	0.63	0.34	0.17	0.10
$\theta 4$ = (0.25, 0.25, 0.25, 0.25)	LH	0.71	0.67	0.57	0.5	0.39	0.20	0.13
	LZL	0.71	0.67	0.57	0.5	0.39	0.20	0.13
	MW	0.71	0.79	0.43	0.63	0.36	0.19	0.14
	TH	0.71	0.67	0.57	0.5	0.39	0.20	0.13
	DY	0.57	0.5	0.57	0.5	0.46	0.23	0.13
	Proposed approach	0.71	0.67	0.57	0.5	0.39	0.20	0.13

Table 7
Fuzzy processing time of tasks for test problem 2.

Task no.	Processing time	Task no.	Processing time
1	(5, 6, 8)	7	(15, 16, 18)
2	(3, 5, 6)	8	(3, 5, 6)
3	(7, 8, 9)	9	(5, 7, 8)
4	(8, 10, 11)	10	(11, 15, 17)
5	(5, 7, 8)	11	(9, 10, 11)
6	(16, 18, 20)	12	(16, 18, 19)

Similar to former test problem, the effect of weight change is assessed on this test problem and the results are presented at Table 8.

As one can see from Table 8, like test problem 1, both LH and LZL methods are not sensitive to weight changes and in many cases, in comparison to other methods, produce worse solutions in term of distance measure. As noted before, the solutions of the proposed approach always dominate DY solutions and this issue holds for solutions provided in Table 8. Also, TH method in comparison with the proposed approach and MW, establishes worse solutions with respect to distance measure. Moreover, there is no

Table 8
Comparison between fuzzy approaches on test problem 2 (δ = 0.01, γ = 0.4).

Weights	Approaches	μ_k				D_1	D_2	D_∞
		μ_1	μ_2	μ_3	μ_4			
$\theta 1$ = (0.45, 0.45, 0.05, 0.05)	LH	0.82	0.61	0.6	0.57	0.3	0.2	0.18
	LZL	0.82	0.61	0.6	0.57	0.3	0.2	0.18
	MW	0.82	0.77	0.25	0.86	0.232	0.14	0.11
	TH	0.82	0.68	0.55	0.64	0.265	0.17	0.14
	DY	0.82	0.77	0.25	0.79	0.235	0.14	0.11
	Proposed approach	0.82	0.77	0.25	0.86	0.232	0.14	0.11
$\theta 2$ = (0.2, 0.6, 0.05, 0.15)	LH	0.82	0.61	0.6	0.57	0.36	0.25	0.24
	LZL	0.82	0.61	0.6	0.57	0.36	0.25	0.24
	MW	0.45	0.98	0.1	0.93	0.18	0.12	0.11
	TH	0.73	0.78	0.5	0.71	0.26	0.15	0.14
	DY	0.36	0.98	0.1	0.93	0.19	0.14	0.13
	Proposed approach	0.45	0.98	0.1	0.93	0.18	0.12	0.11
$\theta 3$ = (0.24, 0.32, 0.18, 0.26)	LH	0.82	0.61	0.6	0.57	0.35	0.19	0.126
	LZL	0.82	0.61	0.6	0.57	0.35	0.19	0.126
	MW	0.73	0.82	0.4	0.79	0.29	0.15	0.108
	TH	0.82	0.68	0.55	0.64	0.32	0.17	0.102
	DY	0.64	0.79	0.45	0.71	0.33	0.17	0.099
	Proposed approach	0.73	0.79	0.45	0.71	0.31	0.15	0.099
$\theta 4$ = (0.25, 0.25, 0.25, 0.25)	LH	0.82	0.61	0.6	0.57	0.35	0.18	0.107
	LZL	0.82	0.61	0.6	0.57	0.35	0.18	0.107
	MW	0.73	0.82	0.4	0.79	0.32	0.18	0.15
	TH	0.82	0.68	0.55	0.64	0.33	0.17	0.112
	DY	0.64	0.59	0.6	0.57	0.40	0.20	0.107
	Proposed approach	0.82	0.61	0.6	0.57	0.35	0.18	0.107

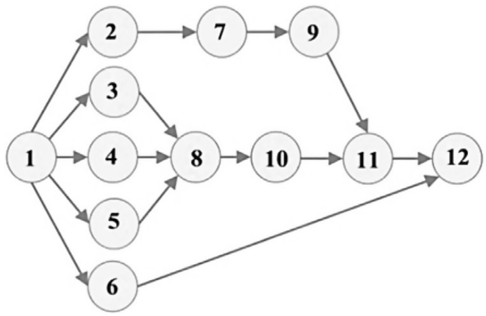

Fig. 8. Precedence diagram of test problem 2.

Table 9
Fuzzy processing time of tasks for test problem 3.

Task no.	Processing time	Task no.	Processing time
1	(5, 6, 7)	9	(4, 5, 6)
2	(4, 5, 6)	10	(3, 4, 5)
3	(1, 2, 3)	11	(5, 6, 7)
4	(8, 9, 10)	12	(4, 5, 6)
5	(7, 8, 9)	13	(5, 6, 7)
6	(3, 4, 5)	14	(3, 4, 5)
7	(6, 7, 8)	15	(2, 3, 4)
8	(3, 4, 5)	16	(3, 4, 5)

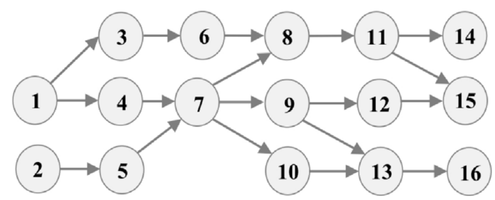

Fig. 9. Precedence diagram of test problem 3.

significant difference between the proposed approach and MW in term of distance measure. However, like the former test problem, in many cases, D_2 and D_∞ take less value for the proposed approach in comparison with MW method which indicates a better balance for satisfaction of objectives.

Further investigations related to the performance of the fuzzy methods in terms of distance measure as well as satisfaction level of objective function along with different weights are provided in Appendix B-Test Problem 2 (Figs. 18–21 and 22–25, respectively).

5.2.3. Test Problem 3

This test problem has presented in [33]. The task processing times for this test problem as well as the precedence constraints are represented in Table 9 and Fig. 9, respectively.

Similar to two previous test problems, the effect of weight change is evaluated on this test problem and the resulted solutions are provided in Table 10.

As shown in Table 10, like two former test problems, both LH and LZL methods are not sensitive to weight change and in many cases in comparison to other methods produce worse solutions with regard to distance measure. Similarly, in this problem, the solutions produced by our proposed approach dominate those which generated by DY method. TH method in comparison with the proposed approach and MW generates worse solutions with respect to distance measure. Moreover, no significant difference exists between proposed approach and MW with regard to distance measure. However, like two former test problems, in the many cases, D_2 and D_∞ take less value for the proposed approach in comparison with MW method which indicates a better balance in satisfaction of objectives.

Further investigations related to the performance of the fuzzy methods in terms of distance measure as well as satisfaction level of the objective function along with different weights are provided in Appendix B-Test Problem 3 (Figs. 26–29 and 30–33, respectively).

Finally, it is concluded that this approach usually leads to find compromise solutions which are more robust and reliable than other approaches in term of distance measure as well as satisfaction level of objectives. Overall, since the introduced approach shows such a great level of performance to deal with

Table 10
Comparison between fuzzy approaches on test problem 3 (δ = 0.01, γ = 0.4).

Weights	Approaches	μ_k				D_1	D_2	D_∞
		μ_1	μ_2	μ_3	μ_4			
$\theta1$ = (0.45, 0.45, 0.05, 0.05)	LH	0.78	0.62	0.53	0.5	0.321	0.202	0.17
	LZL	0.78	0.62	0.53	0.5	0.321	0.202	0.17
	MW	0.78	0.74	0.2	0.79	0.268	0.159	0.12
	TH	0.78	0.68	0.47	0.57	0.290	0.177	0.14
	DY	0.78	0.74	0.2	0.71	0.271	0.160	0.12
	Proposed approach	0.78	0.74	0.2	0.79	0.268	0.159	0.12
$\theta2$ = (0.2, 0.6, 0.05, 0.15)	LH	0.78	0.62	0.53	0.50	0.37	0.25	0.23
	LZL	0.78	0.62	0.53	0.50	0.37	0.25	0.23
	MW	0.44	0.92	0.07	1.00	0.21	0.13	0.11
	TH	0.78	0.68	0.47	0.57	0.32	0.21	0.19
	DY	0.33	0.95	0.13	0.93	0.22	0.14	0.13
	Proposed approach	0.33	0.95	0.13	0.93	0.22	0.14	0.13
$\theta3$ = (0.24, 0.32, 0.18, 0.26)	LH	0.78	0.62	0.53	0.50	0.39	0.20	0.13
	LZL	0.78	0.62	0.53	0.50	0.39	0.20	0.13
	MW	0.56	0.89	0.13	0.93	0.32	0.19	0.16
	TH	0.78	0.68	0.47	0.57	0.36	0.19	0.11
	DY	0.67	0.71	0.40	0.64	0.37	0.19	0.11
	Proposed approach	0.67	0.75	0.40	0.64	0.36	0.18	0.11
$\theta4$ = (0.25, 0.25, 0.25, 0.25)	LH	0.78	0.62	0.53	0.50	0.39	0.20	0.125
	LZL	0.78	0.62	0.53	0.50	0.39	0.20	0.125
	MW	0.78	0.73	0.27	0.79	0.36	0.21	0.183
	TH	0.78	0.62	0.53	0.50	0.39	0.20	0.125
	DY	0.56	0.59	0.53	0.50	0.46	0.23	0.125
	Proposed approach	0.78	0.62	0.53	0.50	0.39	0.20	0.125

imprecise data, as well as its capability to achieve the results that are compatible with the decision maker's preferences; it can be adopted to many practical applications in context of MOLP models.

6. Conclusion

This paper introduces a novel bi-objective fuzzy programming model to formulate the *f*-SALBP as well as the *f*-SULBP. In this line, the proposed model objects to optimize two goals with considering various conflicting objectives.

To do so, a two-phase interactive fuzzy programming approach has been proposed as a solution procedure to solve the problem. In this regard, first, we transform the fuzzy programming model into an auxiliary crisp MOMILP. Then, a new fuzzy approach is employed in order to achieve an efficient compromise solution. After that, several numerical tests have been performed to investigate the validity of the proposed framework. The results show that our new promising interactive fuzzy approach can be used not only for solving this instance problem, but also it would be very helpful to find solution for any MOLP models. Our new FBOMOLP model offers an efficient approach enables the DM to taking into account various confliction objectives as well as interactively adjusting the obtained solution until the efficient compromise solution which is compatible with the DM's point of view is derived. At last, although the attention of this study was paid to develop a single-model of ALBP but it is hoped that this direction can be followed by developing much complex problems such as Mixed-Model assembly lines in the future.

Appendix A.

Herein, an abstract version of five methods which are considered in this paper (i.e. the LH, LZL, MW, DY, and TH) is provided.

Li et al. (LZL) two-phase method [53]

$$\begin{aligned}&\max \lambda(x)=\sum_{k=1}^{K}\theta_k\mu_k(x)\\&S.t:\lambda_k^0\le\mu_k(x),\quad k=1\ldots K\\&x\in F(x)\\&\mu_k(x)\ \text{ and }\ \lambda_0^k\in[0,1],\quad k=1,\ldots,K\end{aligned}\tag{47}$$

where, λ_k^0 refers to the minimum satisfaction degree for *k*th objective function, which can be achieved use of max–min approach suggested by Zimmermann [51], as (presented in the following) follows:

$$\begin{aligned}&\max\{\lambda\}\\&S.t:\lambda\le\mu_k(x),\quad k=1,\ldots,K\\&x\in FS(x)\\&\lambda\in[0,1],\quad k=1,\ldots,K\end{aligned}\tag{48}$$

Demirli and Yimer (DY) weighted max–min method [57]

$$\begin{aligned}&\max\{\lambda\}\\&S.t:\theta_k\lambda\le\mu_k(x),\quad k=1,\ldots,K\\&x\in FS(x)\\&\lambda\in[0,1],\quad k=1,\ldots,K\end{aligned}\tag{49}$$

where θ_k represents the *k*th objective weight which is defined by the DM with respect to his/her preference.

Lai and Hwang (LH) augmented max–min method [52]

$$\begin{aligned}&\max \lambda(x)=\lambda_0+\delta\sum_{k=1}^{K}\theta_k\mu_k(x)\\&S.t:\lambda_0\le\mu_k(x),\quad k=1,\ldots,K\\&x\in F(x)\\&\mu_k(x)\ \text{ and }\ \lambda_0\in[0,1],\quad k=1,\ldots,K\end{aligned}\tag{50}$$

In the above formulation, $\mu_k(x)$ and λ_0 denote the satisfaction degree of *k*th objective and minimum satisfaction degree of objectives, respectively. Also, δ is referred to a positive number which is sufficiently small as it usually set to 0.01.

Selim and Ozkarahan modified Werners (MW) method [54]

$$\begin{aligned}&\max \lambda(x)=\gamma\lambda_0+(1-\gamma)\sum_{k=1}^{K}\theta_k\lambda_k\\&S.t:\lambda_0+\lambda_k\le\mu_k(x),\quad k=1,\ldots,K\\&x\in F(x)\\&\gamma,\lambda_0,\lambda_k\in[0,1],\quad k=1,\ldots,K\end{aligned}\tag{51}$$

In this model, $\mu_k(x)$ indicates the level of satisfaction and λ_0 represents the minimum level of satisfaction. Also, γ is related to the coefficient of compensation which controls the compromise degree among the objectives as well as the minimum satisfaction level of objectives. In this study, according to the initial test performed in [56], we have set it to 0.4.

Torabi and Hassini hybrid method [56]

$$\begin{aligned}&\max \lambda(x)=\gamma\lambda_0+(1-\gamma)\sum_{k=1}^{K}\theta_k\mu_k\\&S.t:\lambda_0\le\mu_k(x),\quad k=1,\ldots,K\\&x\in F(x)\\&\gamma,\lambda_0\in[0,1],\quad k=1\ldots K\end{aligned}\tag{52}$$

Appendix B.

Test Problem 1

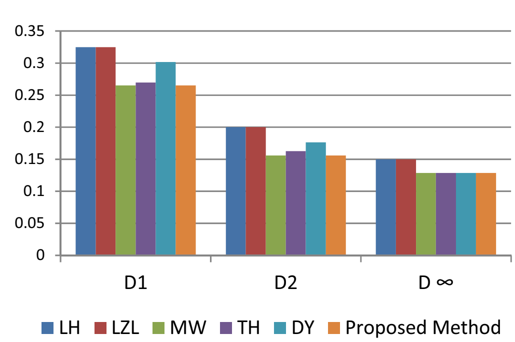

Fig. 10. Distance measures for existing approaches on the test problem 1 (θ = 0.45, 0.45, 0.05, 0.05).

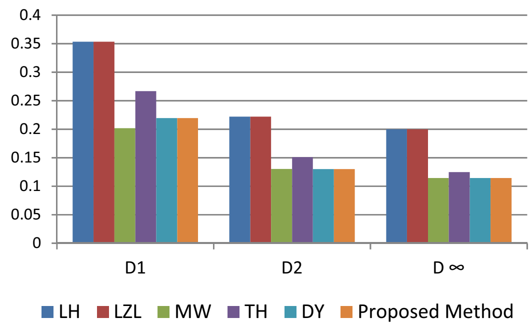

Fig. 11. Distance measures for existing approaches on the test problem 1 ($\theta = 0.2$, 0.6, 0.05, 0.15).

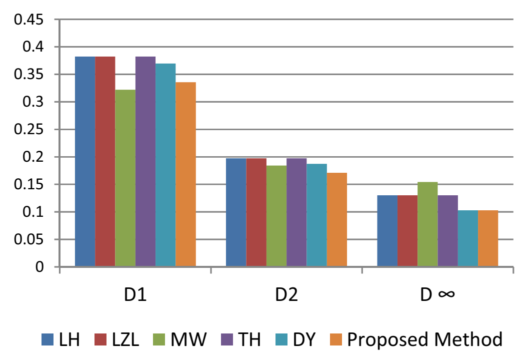

Fig. 12. Distance measures for existing approaches on the test problem 1 ($\theta = 0.24$, 0.32, 0.18, 0.26).

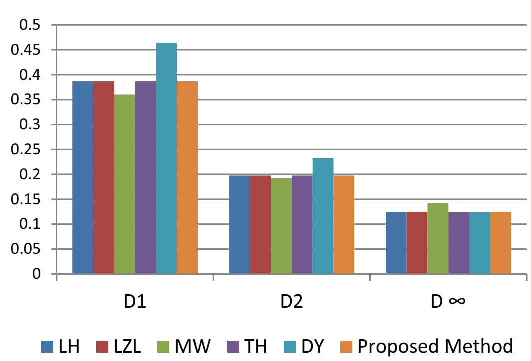

Fig. 13. Distance measures for existing approaches on the test problem 1 ($\theta = 0.25$, 0.25, 0.25, 0.25).

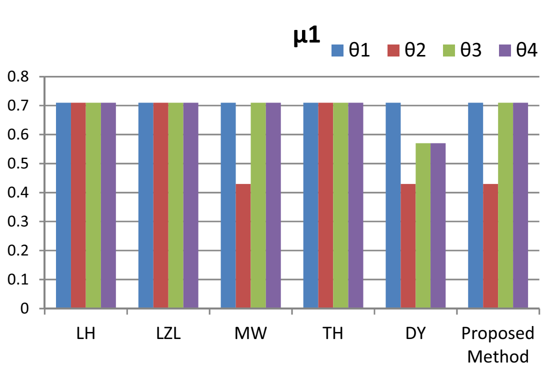

Fig. 14. Satisfaction level of μ_1 for different weights on the test problem 1.

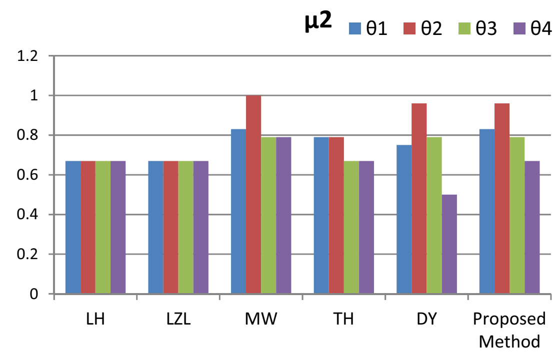

Fig. 15. Satisfaction level of μ_2 for different weights on the test problem 1.

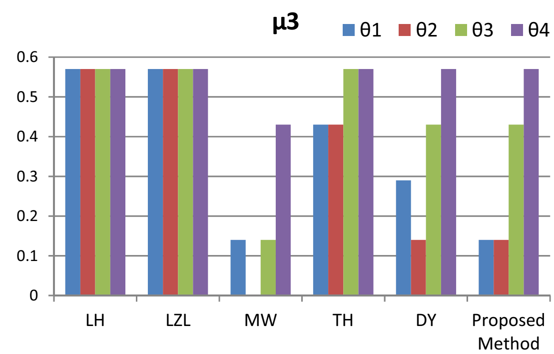

Fig. 16. Satisfaction level of μ_3 for different weights on the test problem 1.

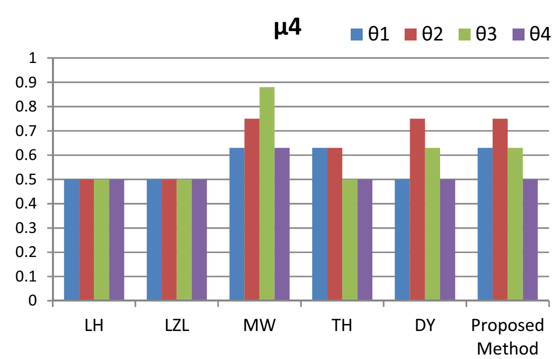

Fig. 17. Satisfaction level of μ_4 for different weights on the test problem 1.

Test Problem 2

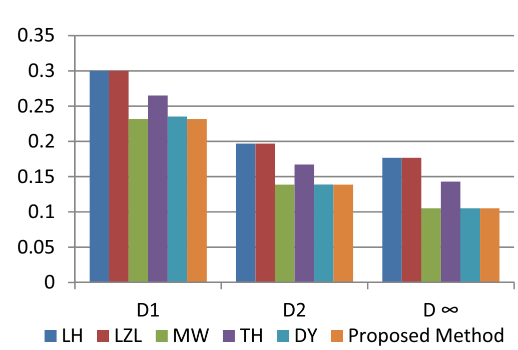

Fig. 18. Distance measures for existing approaches on the test problem 2 ($\theta = 0.45$, 0.45, 0.05, 0.05).

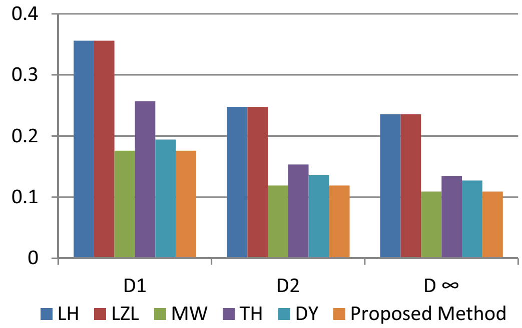

Fig. 19. Distance measures for existing approaches on the test problem 2 (θ = 0.2, 0.6, 0.05, 0.15).

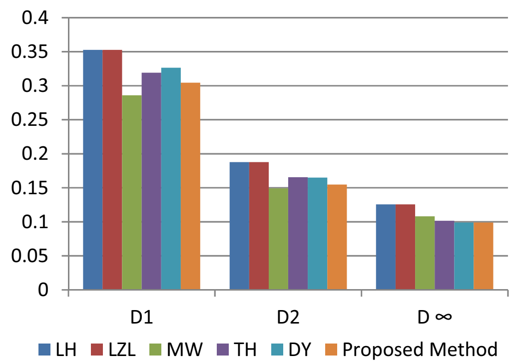

Fig. 20. Distance measures for existing approaches on the test problem 2 (θ = 0.24, 0.32, 0.18, 0.26).

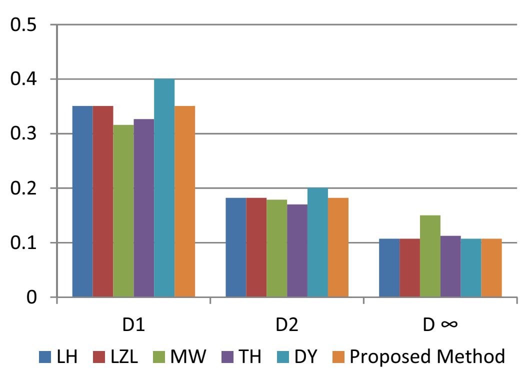

Fig. 21. Distance measures for existing approaches on the test problem 2 (θ = 0.25, 0.25, 0.25, 0.25).

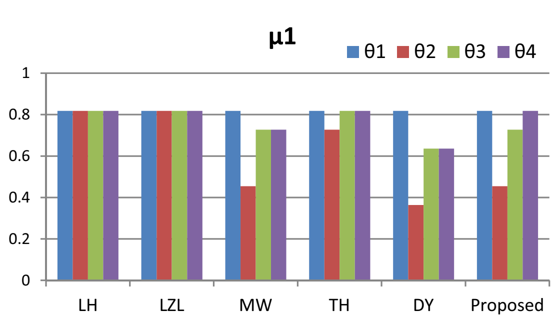

Fig. 22. Satisfaction level of μ_1 for different weights on the test problem 2.

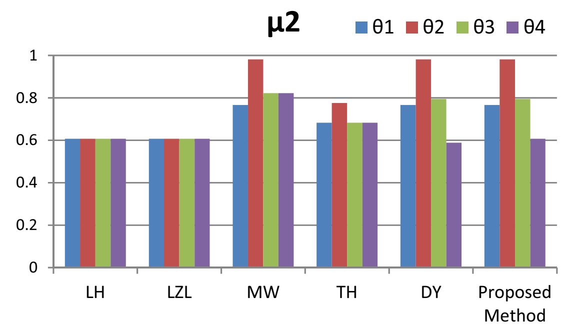

Fig. 23. Satisfaction level of μ_2 for different weights on the test problem 2.

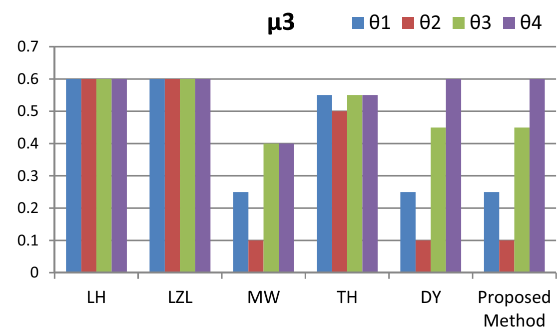

Fig. 24. Satisfaction level of μ_3 for different weights on the test problem 2.

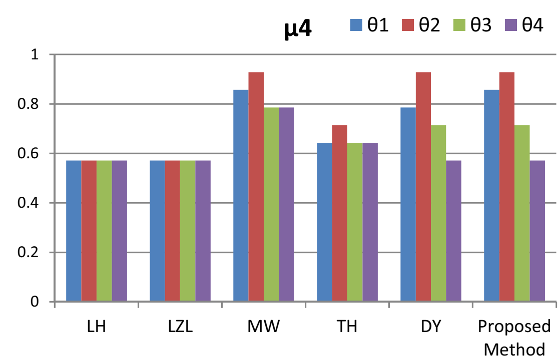

Fig. 25. Satisfaction level of μ_4 for different weights on the test problem 2.

Test Problem 3

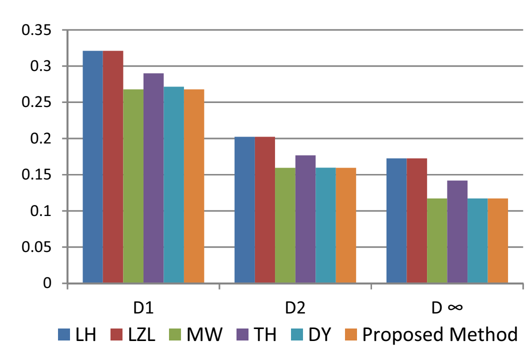

Fig. 26. Distance measures for existing approaches on the test problem 3 (θ = 0.45, 0.45, 0.05, 0.05).

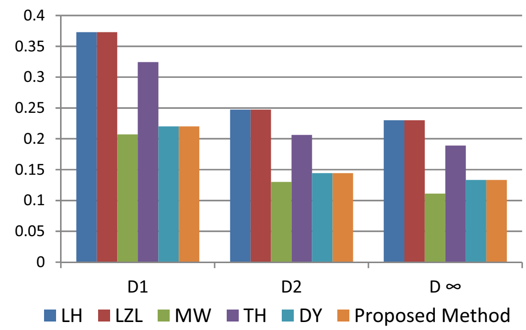

Fig. 27. Distance measures for existing approaches on the test problem 3 (θ = 0.2, 0.6, 0.05, 0.15).

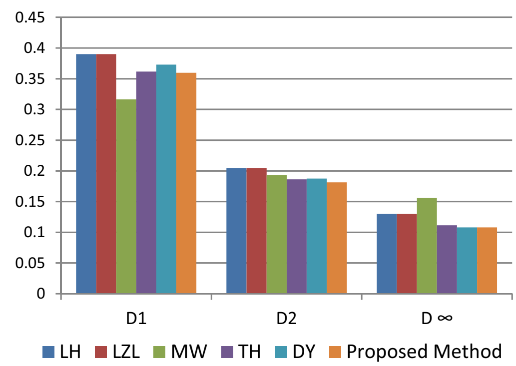

Fig. 28. Distance measures for existing approaches on the test problem 3 (θ = 0.24, 0.32, 0.18, 0.26).

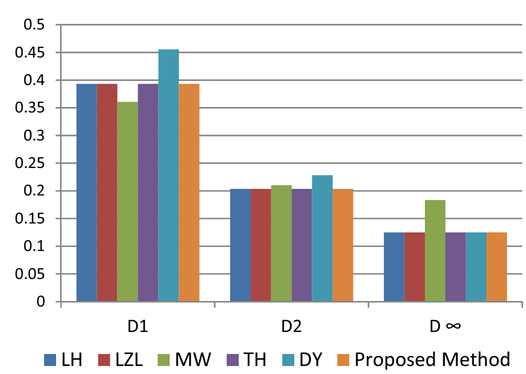

Fig. 29. Distance measures for existing approaches on the test problem 3 (θ = 0.25, 0.25, 0.25, 0.25).

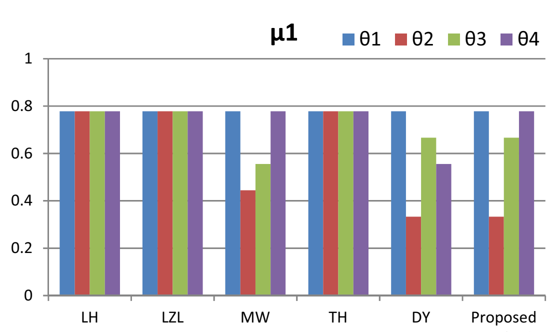

Fig. 30. Satisfaction level of μ_1 for different weights on the test problem 3.

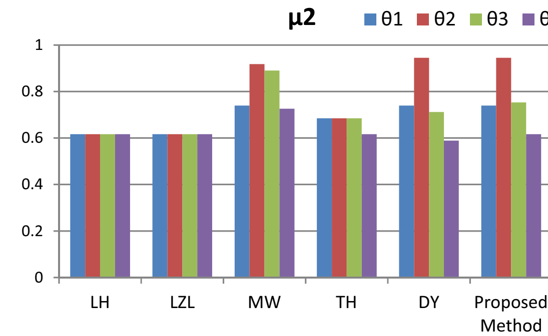

Fig. 31. Satisfaction level of μ_2 for different weights on the test problem 3.

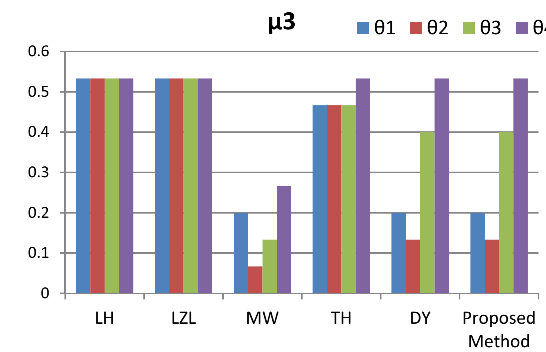

Fig. 32. Satisfaction level of μ_3 for different weights on the test problem 3.

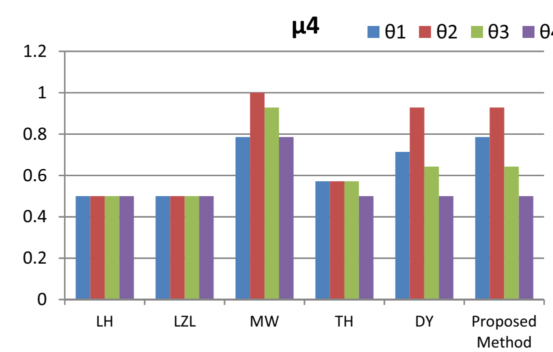

Fig. 33. Satisfaction level of μ_4 for different weights on the test problem 3.

References

[1] M. Baudin, Lean Assembly: The Nuts and Bolts of Making Assembly Operations Flow, Productivity Press, New York, USA, 2002.
[2] E.P.C. Kao, A preference order dynamic program for stochastic assembly line balancing, Manag. Sci. 22 (1976) 1097–1104.
[3] C. Becker, A. Scholl, A survey on problems and methods in generalized assembly line balancing, Eur. J. Oper. Res. 168 (2006) 694–715.
[4] A. Scholl, C. Becker, State-of-the-art exact and heuristic solution procedures for simple assembly line balancing, Eur. J. Oper. Res. 168 (2006) 666–693.
[5] Y. Monden, Toyota Production System: An Integrated Approach to Just-In-Time, CRC Press, 2012.
[6] M.E. Salveson, The assembly line balancing problem, J. Ind. Eng. 6 (1955) 18–25.
[7] I. Baybars, A survey of exact algorithms for the simple assembly line balancing problem, Manag. Sci. 32 (1986) 909–932.
[8] S. Ghosh, R.J. Gagnon, A comprehensive literature review and analysis of the design, balancing and scheduling of assembly systems, Int. J. Prod. Res. 27 (1989) 637–670.
[9] E. Erel, S.C. Sarin, A survey of the assembly line balancing procedures, Prod. Plan. Control 9 (1998) 414–434.
[10] S. Tasan, S. Tunali, A review of the current applications of genetic algorithms in assembly line balancing, J. Intell. Manuf. 19 (2008) 49–69.
[11] D.M.K. Naveen, Assembly Line Balancing: A Review of Developments and Trends in Approach to Industrial Application, 2013.

[12] K. Ağpak, S. Zolfaghari, Mathematical models for parallel two-sided assembly line balancing problems and extensions, Int. J. Prod. Res. 53 (2015) 1242–1254.
[13] S.K. Gupta, Assembly Line Balancing and Optimization through Mathematical Modeling, 2015.
[14] M.C.O. Moreira, C. Miralles, A.M. Costa, Model and heuristics for the assembly line worker integration and balancing problem, Comput. Oper. Res. 54 (2015) 64–73.
[15] A. Roshani, D. Giglio, A Mathematical Programming Formulation for Cost-oriented Multi-manned Assembly Line Balancing Problem, 2015.
[16] H. Triki, W. Hachicha, A. Mellouli, F. Masmoudi, An assembly line balancing problem automotive cables, Manag. Prod. Eng. Rev. 6 (2015) 59–66.
[17] S. Choudhary, S. Agrawal, Multiobjective mixed model assembly line balancing problem, in: E.S. M. El-Alfy, S.M. Thampi, H. Takagi, S. Piramuthu, T. Hanne (Eds.), Advances in Intelligent Informatics, vol. 320, Springer International Publishing, 2015, pp. 655–663.
[18] J. Pereira, Empirical evaluation of lower bounding methods for the simple assembly line balancing problem, International Journal of Production Research 53 (2015) 3327–3340.
[19] G. Miltenburg, J. Wijngaard, U-line line balancing problem, Manag. Sci. 40 (1994) 1378–1388.
[20] T.L. Urban, Note. Optimal balancing of U-shaped assembly lines, Manag. Sci. 44 (1998) 738–741.
[21] H. Gökçen, K. Ağpak, A goal programming approach to simple U-line balancing problem, Eur. J. Oper. Res. 171 (2006) 577–585.
[22] R.F. Deckro, S. Rangachari, A goal approach to assembly line balancing, Comput. Oper. Res. 17 (1990) 509–521.
[23] A. Fattahi, M. Turkay, On the MILP model for the U-shaped assembly line balancing problems, Eur. J. Oper. Res. 242 (2015) 343–346.
[24] A. Fattahi, S. Elaoud, E. Sadeqi Azer, M. Turkay, A novel integer programming formulation with logic cuts for the U-shaped assembly line balancing problem, Int. J. Prod. Res. 52 (2014) 1318–1333.
[25] P.T. Zacharia, A.C. Nearchou, Multi-objective fuzzy assembly line balancing using genetic algorithms, J. Intell. Manuf. 23 (2012) 615–627.
[26] L.A. Zadeh, Fuzzy sets, Inf. Control 8 (1965) 338–353.
[27] Y. Tsujimura, M. Gen, E. Kubota, Solving fuzzy assembly-line balancing problem with genetic algorithms, Comput. Ind. Eng. 29 (1995) 543–547.
[28] M. Gen, Y. Tsujimura, Y. Li, Fuzzy assembly line balancing using genetic algorithms, Comput. Ind. Eng. 31 (1996) 631–634.
[29] M.H. Alavidoost, M.H.F. Zarandi, M. Tarimoradi, Y. Nemati, Modified genetic algorithm for simple straight and U-shaped assembly line balancing with fuzzy processing times, J. Intell. Manuf. (2014) 1–24.
[30] M.H. Alavidoost, M. Tarimoradi, M.H.F. Zarandi, Fuzzy adaptive genetic algorithm for multi-objective assembly line balancing problems, Appl. Soft Comput. 34 (2015) 655–677.
[31] M.H. Alavidoost, M.A. Nayeri, Proposition of a hybrid NSGA-II algorithm with fuzzy objectives for bi-objective assembly line balancing problem, in: Tenth International Industrial Engineering Conference, 2014.
[32] P.T. Zacharia, A.C. Nearchou, A meta-heuristic algorithm for the fuzzy assembly line balancing type-E problem, Comput. Oper. Res. 40 (2013) 3033–3044.
[33] D. Fonseca, C. Guest, M. Elam, C. Karr, A fuzzy logic approach to assembly line balancing, Mathw. Soft Comput. 12 (2005) 57–74.
[34] M. Tarimoradi, M.H. Alavidoost, M.H.F. Zarandi, Comparative Corrigendum note on papers "Fuzzy adaptive GA for multi-objective assembly line balancing" continued "Modified GA for different types of assembly line balancing with fuzzy processing times": differences and similarities [Appl. Soft Comput. 34 (September 2015) 655–677], Appl. Soft Comput. 35 (2015) 786–788.
[35] R. Behbamzadeh, M. Alaghebandha, A. Azizi, A mixed-integer model with genetic algorithm for multi-objective assembly line balancing problem in fuzzy manufacturing environment, IJEIR 4 (2015) 172–178.
[36] Z. Zhang, W. Cheng, Improved heuristic procedure for mixed-model u-line balancing problem with fuzzy times, in: Proceedings of China Modern Logistics Engineering, Springer, 2015, pp. 395–406.
[37] W. Zhang, W. Xu, G. Liu, M. Gen, An effective hybrid evolutionary algorithm for stochastic multiobjective assembly line balancing problem, J. Intell. Manuf. (2015) 1–8.
[38] W. Zhang, W. Xu, M. Gen, Hybrid multiobjective evolutionary algorithm for assembly line balancing problem with stochastic processing time, Procedia Comput. Sci. 36 (2014) 587–592.
[39] W.-C. Chiang, T.L. Urban, C. Luo, Balancing stochastic two-sided assembly lines, Int. J. Prod. Res. (2015) 1–19.
[40] G. La Scalia, Solving type-2 assembly line balancing problem with fuzzy binary linear programming, J. Intell. Fuzzy Syst. 25 (2013) 517–524.
[41] Z.Q. Zhang, W.M. Cheng, Solving fuzzy u-shaped line balancing problem with exact method, Appl. Mech. Mater. 26 (2010) 1046–1051.
[42] B. Toklu, U. özcan, A fuzzy goal programming model for the simple U-line balancing problem with multiple objectives, Eng. Optim. 40 (2008) 191–204.
[43] Y. Kara, T. Paksoy, C.T. Chang, Binary fuzzy goal programming approach to single model straight and U-shaped assembly line balancing, Eur. J. Oper. Res. 195 (2009) 335–347.
[44] Ö. Hazır, A. Dolgui, A decomposition based solution algorithm for U-type assembly line balancing with interval data, Comput. Oper. Res. 59 (2015) 126–131.
[45] M. Zarandi, M. Tarimoradi, M. Alavidoost, B. Shakeri, Fuzzy approximate reasoning toward multi-objective optimization policy: deployment for supply chain programming, in: Fuzzy Information Processing Society (NAFIPS) held jointly with 2015 5th World Conference on Soft Computing (WConSC), 2015 Annual Conference of the North American, 2015, pp. 1–6.
[46] M. Zarandi, M. Tarimoradi, M. Alavidoost, M. Shirazi, Fuzzy comparison dashboard for multi-objective evolutionary applications: an implementation in supply chain planning, in: Fuzzy Information Processing Society (NAFIPS) held jointly with 2015 5th World Conference on Soft Computing (WConSC), 2015 Annual Conference of the North American, 2015, pp. 1–6.
[47] Y.-J. Lai, C.-L. Hwang, A new approach to some possibilistic linear programming problems, Fuzzy Sets Syst. 49 (1992) 121–133.
[48] R.-C. Wang, T.-F. Liang, Applying possibilistic linear programming to aggregate production planning, Int. J. Prod. Econ. 98 (2005) 328–341.
[49] M.H. Alavidoost, M. Tarimoradi, M.H.F. Zarandi, Bi-objective mixed-integer nonlinear programming for multi-commodity tri-echelon supply chain networks, J. Intell. Manuf. (2015) 1–18.
[50] R.E. Bellman, L.A. Zadeh, Decision-making in a fuzzy environment, Manag. Sci. 17 (1970) B141–B164.
[51] H.-J. Zimmermann, Fuzzy programming and linear programming with several objective functions, Fuzzy Sets Syst. 1 (1978) 45–55.
[52] Y.-J. Lai, C.-L. Hwang, Possibilistic linear programming for managing interest rate risk, Fuzzy Sets Syst. 54 (1993) 135–146.
[53] X.-q. Li, B. Zhang, H. Li, Computing efficient solutions to fuzzy multiple objective linear programming problems, Fuzzy Sets Syst. 157 (2006) 1328–1332.
[54] H. Selim, I. Ozkarahan, A supply chain distribution network design model: an interactive fuzzy goal programming-based solution approach, Int. J. Adv. Manuf. Technol. 36 (2008) 401–418.
[55] B.M. Werners, Aggregation models in mathematical programming, in: Springer (Ed.), Mathematical Models for Decision Support, 1988, pp. 295–305.
[56] S.A. Torabi, E. Hassini, An interactive possibilistic programming approach for multiple objective supply chain master planning, Fuzzy Sets Syst. 159 (2008) 193–214.
[57] K. Demirli, A.D. Yimer, Fuzzy scheduling of a build-to-order supply chain, Int. J. Prod. Res. 46 (2008) 3931–3958.
[58] W.F. Abd El-Wahed, S.M. Lee, Interactive fuzzy goal programming for multi-objective transportation problems, Omega 34 (2006) 158–166.
[59] Y.-J. Lai, C.-L. Hwang, Fuzzy mathematical programming. Methods and applications, Lect. Notes Econ. Math. Syst. 394 (1992).